MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

END OF SEMESTER EXAMINATION

ICT 4701: SOFTWARE ENGINEERING

TIME ALLOWED: 3 HOURS **DATE : 6th AUGUST, 2018**

INSTRUCTIONS

i. This paper contains **Two** pages. Please check.
ii. There are **Three questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

Question one

a) Describe the different consumers of software requirements i.e. the different users, or types of users, of a software requirements document. For each consumer, explain how he or she would use the requirements.
(10 marks)

b) One source of problems in the requirements phase can be the relationship between system developers and their customers. What are some negative stereotypes customers may hold about developers? What could you, as a developer on a project, do to minimize the impact of those negative stereotypes? **(10 marks)**

c) Describe the differences between acceptance and beta testing. Give the goals for each type of testing and describe when and how each should occur. **(10 marks)**

d) Explain why interface testing is necessary even when individual components have been extensively validated through component testing and program inspections. **(10 marks)**

Question two

a) Giving reasons for your answer based in the type of system being developed, suggest the most appropriate generic software process model that might be used as basis for managing the development of the following system:

i. A system to control anti-locking in a car. **(2 marks)**

ii. A virtual reality system to support software maintenance.
(2 marks)

iii. A university accounting system that replaces an existing system.
(2 marks)

iv. An interactive system that allows railway passengers to find train times from terminals installed in stations. **(2 marks)**

Question three

a) Distinguish between functional requirements and non-functional requirements. **(10 marks)**

b) Discuss **five** problems which might be encountered during requirements engineering/analysis. **(10 marks)**

c) You have been hired by Akatswiri Softwares Inc as a Systems Analyst to develop a computerized system for Tikondane banki mkhonde group. Your immediate supervisor has requested you to solicit requirements and design the system.

i. Draw a use case diagram **(12 marks)**

ii. Draw an entity relationship diagram **(20 marks)**

END OF QUESTION PAPER